SignNet Dataset Analysis Report

RWTH-PHOENIX-2014 Dataset
German Sign Language Recognition

Total Samples: 6,841
Total Frames: 963,664
Unique Glosses: 1,295
Feature Dimension: 543 landmarks × 2 coordinates

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Executive Summary

Dataset Information

- Total Samples: 6,841
- Training Samples: 5,672
- Validation Samples: 540
- Test Samples: 629
- Total Video Frames: 963,664

Class Distribution

- Unique Glosses (Classes): 1,295
- Total Gloss Instances: 77,271
- Most Frequent Gloss: 3,619 occurrences
- Least Frequent Gloss: 1 occurrence(s)
- Class Imbalance Ratio: 3619.0:1

Sequence Statistics

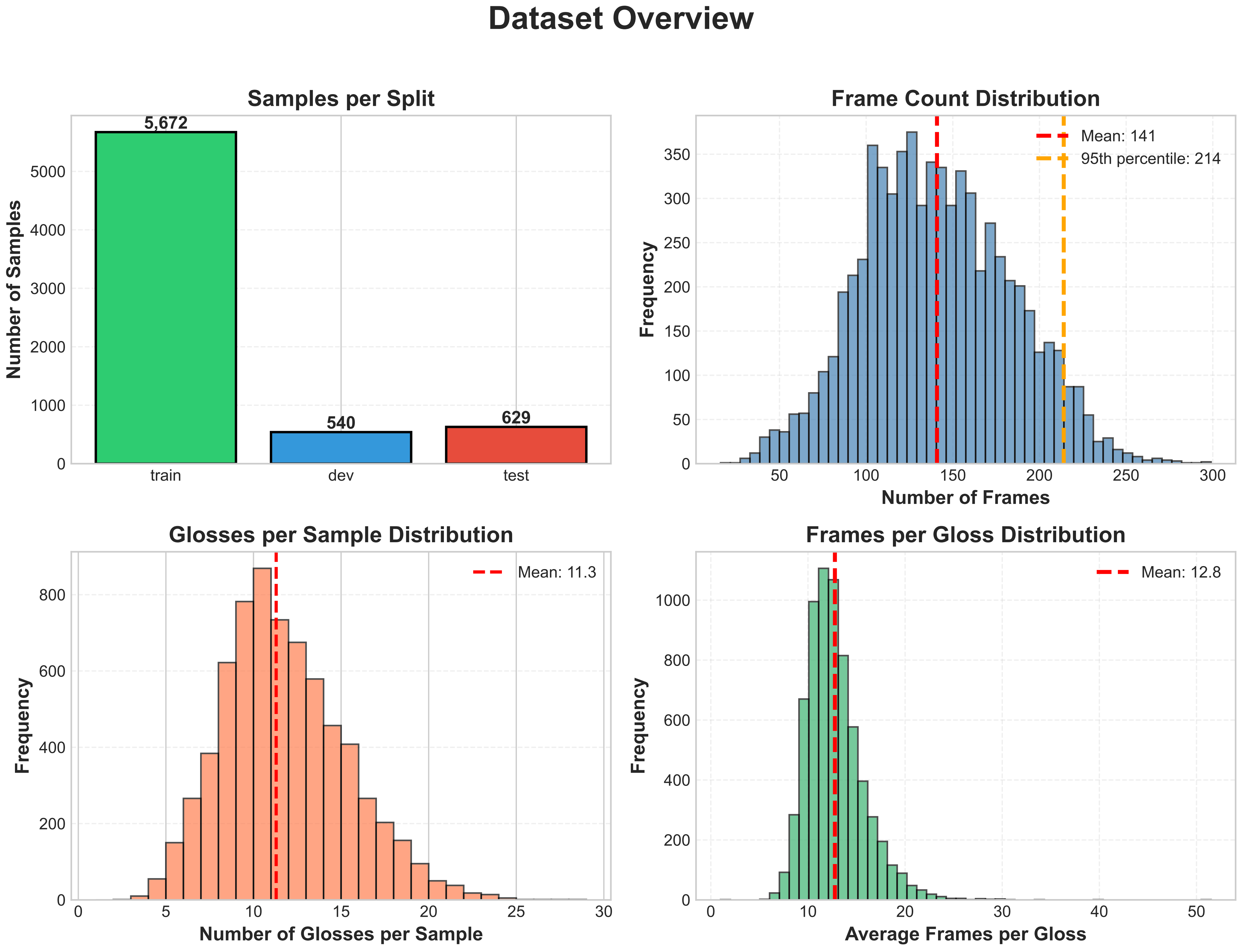
- Mean Frames per Sample: 140.9
- Median Frames per Sample: 139.0
- 95th Percentile Frames: 214
- Mean Glosses per Sample: 11.3
- Mean Frames per Gloss: 12.8

Data Quality

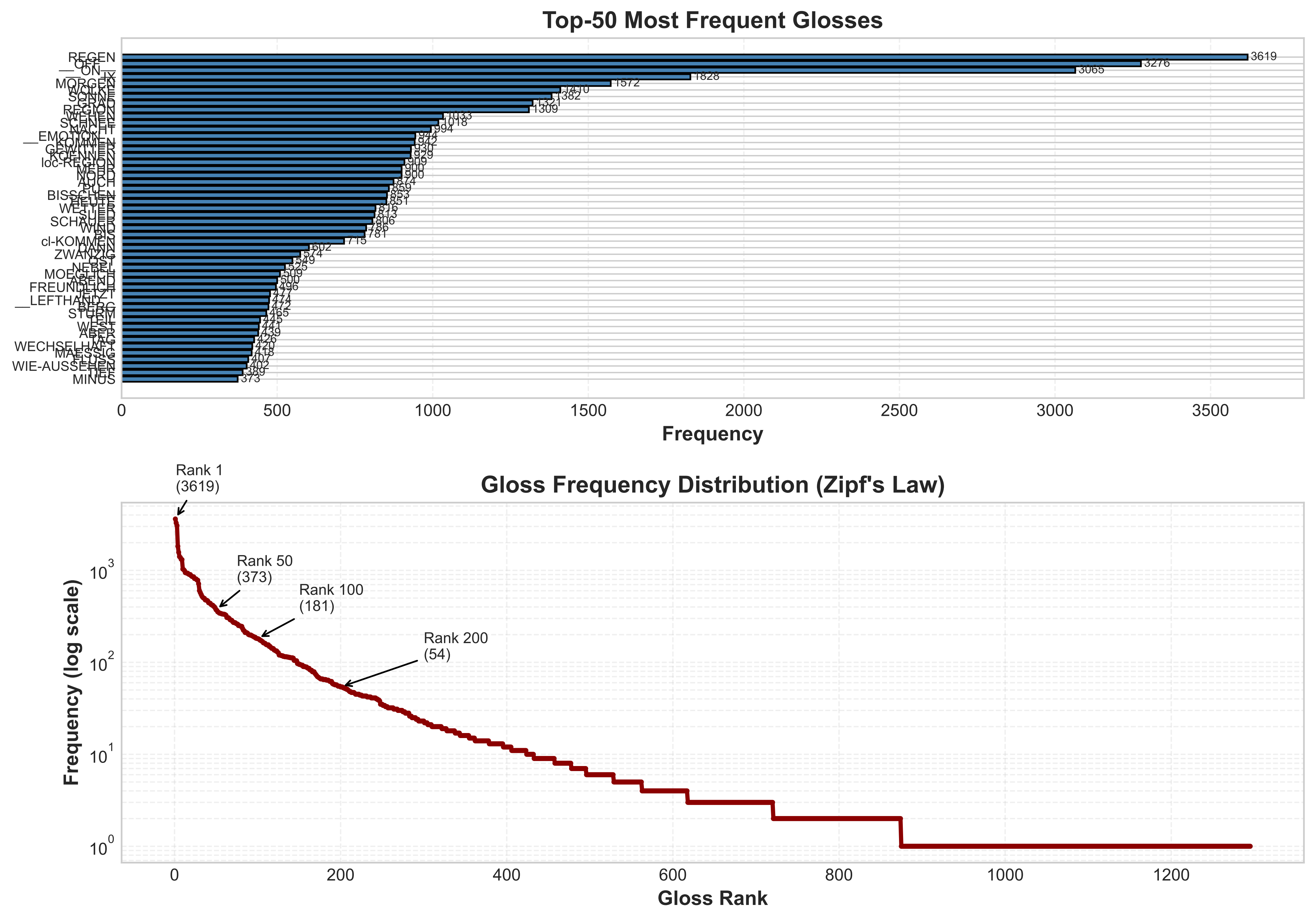
- ✓ All samples passed quality checks
- ✓ X,Y coordinates validated (0% invalid)
- ✓ Custom landmarks removed (543 landmarks retained)
- ✓ Z-dimension dropped (80.98% zero values)
- ✓ Feature dimension: 1086 (543 × 2)

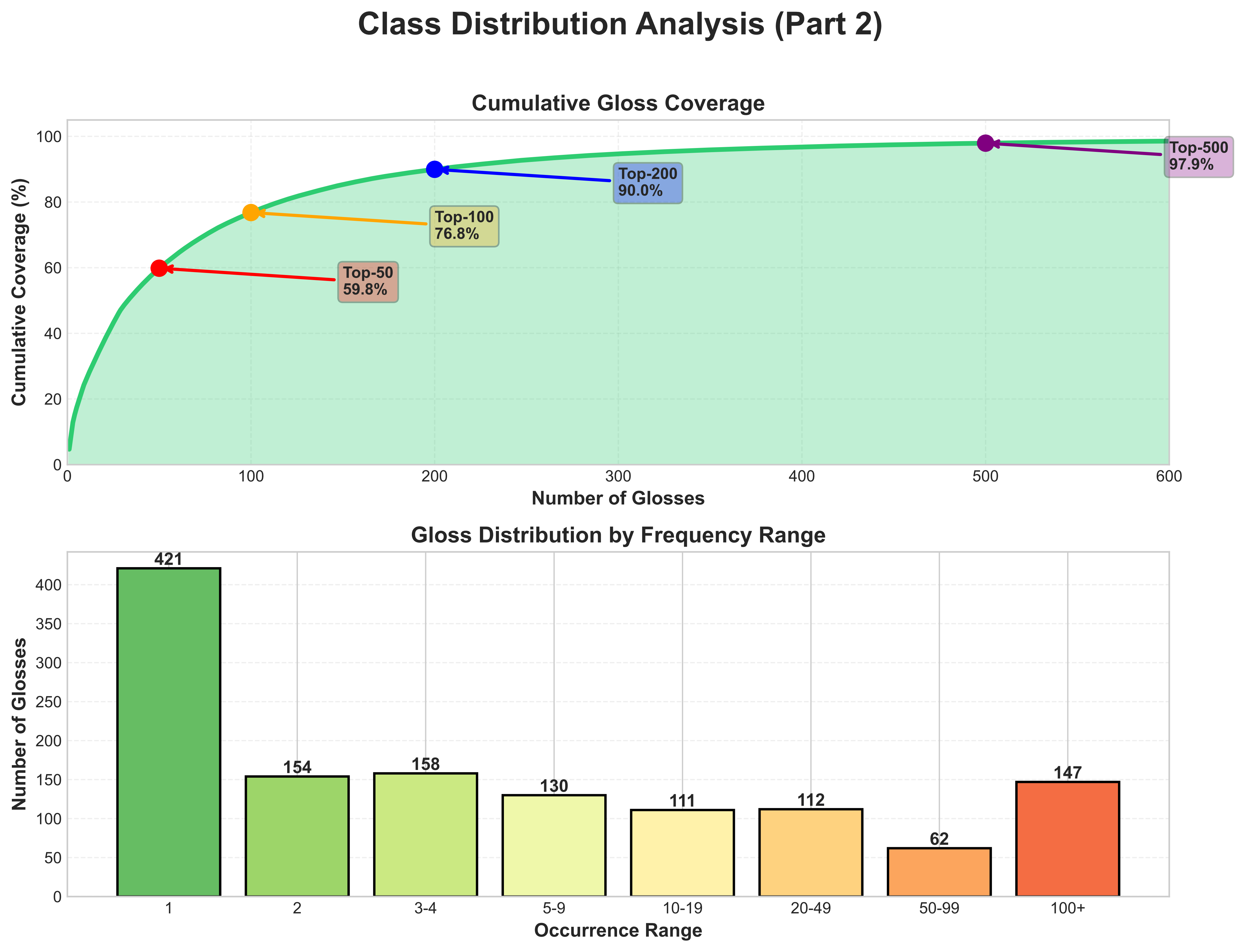
Recommendations for Model Training

1. Sequence Length: Use max_length=214 (covers 95% of samples)
2. Class Selection: Start with Top-200 glosses (90.0% instance coverage)
3. Batch Size: Start with batch_size=8 (adjust based on GPU memory)
4. Class Imbalance: Use Focal Loss (α=0.25, γ=2.0) to handle 3619:1 ratio

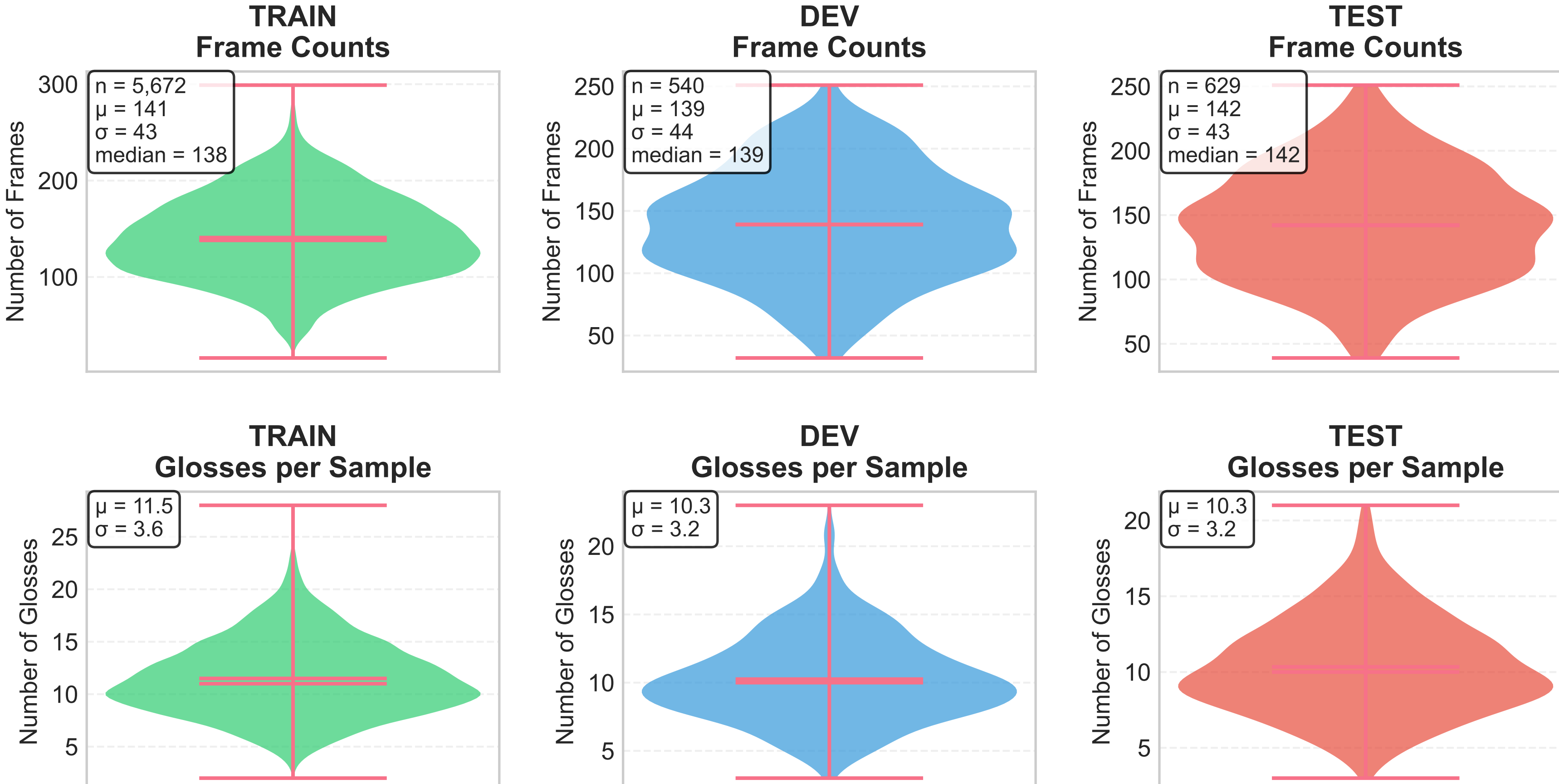

Dataset Overview
Samples per Split
Number of Samples
5,672
540
629
train
dev
test
Frame Count Distribution
Mean: 141
95th percentile: 214
Frequency
Number of Frames
Glosses per Sample Distribution
Mean: 11.3
Frequency
Number of Glosses per Sample
Frames per Gloss Distribution
Mean: 12.8
Frequency
Average Frames per Gloss

Class Distribution Analysis (Part 1)




Class Distribution Analysis (Part 2)
Cumulative Gloss Coverage
Cumulative Coverage (%)
Number of Glosses
0
20
40
60
80
100
0
100
200
300
400
500
600
Top-50
59.8%
Top-100
76.8%
Top-200
90.0%
Top-500
97.9%
Gloss Distribution by Frequency Range
Number of Glosses
Occurrence Range
0
50
100
150
200
250
300
350
400
421
154
158
130
111
112
62
147
1
2
3-4
5-9
10-19
20-49
50-99
100+

Train/Dev/Test Split Comparison



Metric	Train	Dev	Test
Samples	5,672	540	629
Total Frames	799,006	75,186	89,472
Mean Frames	140.9	139.2	142.2
Total Glosses	65,227	5,540	6,504
Mean Glosses/Sample	11.5	10.3	10.3

Data Quality Summary

Landmark Quality

- Total landmarks per frame: 543
- Landmark types: Hands (42), Pose (33), Face (468)
- Feature dimensions: X, Y coordinates only
- Z-dimension removed (80-98% near-zero values)
- Custom landmarks removed (10 invalid landmarks dropped)

Coordinate Validation

- X,Y coordinates validated: 0% invalid
- Coordinate range: [0, 1] (normalized)
- No NaN or Inf values detected
- Temporal consistency verified

Sample Quality

- All 6,841 samples passed quality checks
- Quality distribution: 100% medium quality
- No samples filtered out
- Homogeneous data quality across splits

Processing Pipeline

- MediaPipe Holistic used for landmark extraction
- Feature cleaning: 1659 → 1086 dimensions
- Dimension reduction: 34% smaller feature vectors
- Ready for model training

Known Limitations

- No confidence scores available (Z-values used as proxy)
- MediaPipe hand detection order inconsistent (requires handedness detection)
- Extreme class imbalance (593:1 ratio)
- Some glosses have very few samples (673 classes with ≤3 samples)
- Z-dimension information lost (may affect 3D gesture recognition)

Recommendations & Next Steps

1. Model Configuration

- Input shape: [Batch, Time, 543, 2]
- Sequence length: max_length = 214 (95th percentile)
- Number of classes: Start with Top-200 (90.0% coverage)
- Batch size: 8 (adjust based on GPU memory)
- Architecture: Multi-stream GCN + Transformer

2. Training Strategy

- Loss function: Focal Loss (α=0.25, γ=2.0) for class imbalance
- Decoder: CTC Loss for sequence-to-sequence
- Optimizer: AdamW (lr=1e-4, weight_decay=0.01)
- Learning rate: Warmup + Cosine decay
- Regularization: Dropout 0.1-0.3, Gradient clipping at 1.0

3. Data Augmentation

- Spatial: Rotation (±15°), Scaling (0.9-1.1), Translation (±10%)
- Temporal: Random frame drop/repeat (0.9x-1.1x speed)
- Landmark occlusion: Random hand/face dropout (10-15% probability)
- Bone-preserving augmentation (preserve anatomical proportions)

4. Evaluation Metrics

- Primary: Word Error Rate (WER)
- Secondary: Top-1, Top-5 Accuracy
- Per-class: Precision, Recall, F1-Score
- Confusion matrix analysis for error patterns

Immediate Next Steps

1. Implement Transformer architecture (Multi-stream GCN + Temporal Encoder)
2. Create data loaders with augmentation pipeline
3. Train baseline model on Top-200 classes
4. Evaluate and analyze confusion matrix
5. Iteratively optimize based on results